



Knowledge check-based concept mapping in digital games: impacts on students' learning performance and behaviors

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Abstract

Many studies have incorporated concept maps into digital games to enable learners to make connections between subject concepts in the game. However, most learners do not associate spontaneously with the thematic concepts in the game but need to be facilitated by effective scaffolding mechanisms to reconceptualize the learning process and content. Therefore, this study proposed a knowledge check-based concept map-guided gaming strategy for chemistry learning for secondary-level learners. Specifically, the knowledge check-based concept map-guided gaming strategy was employed as a learning strategy while playing digital games. The knowledge check mechanism was developed based on the notation of two-tier testing. Moreover, the learning environment was a self-developed chemical classification-based digital game, in which learners learned through concept mapping guided gaming tasks with factual and conceptual questions to confirm that they can actually learn content and conceptual understanding in the game. To understand the effects of the proposed strategy, a quasi-experimental design was adopted to measure compare two groups' learning achievement, problem-solving awareness, critical thinking tendency, and germane cognitive load of seventh-grade students using the knowledge check-based concept map-guided gaming strategy and the conventional concept map-guided gaming strategy. The results of the study showed that learners' learning achievement, problem-solving awareness, critical thinking tendency, and germane cognitive load in chemistry learning were effectively improved by the knowledge check-based concept map-guided gaming strategy. Furthermore, this study implies that teachers can adjust their teaching scaffolds and provide strategies according to learners' learning behaviors in digital game-based learning.

Keywords Knowledge check · Concept map · Chemistry · Game-based learning

Introduction

Chemistry is essential to understanding most areas of science, technology, and engineering; technology innovators need to understand chemistry well (Balaban & Klein, 2006; Limniou et al., 2008). Chemistry is often referred to as the central science (Brown et al., 2021a, 2021b; Chang, 2007) as it includes many abstract concepts essential for further study of

other sciences (Brown et al., 2021a, 2021b). Chemistry studies the properties, attributes, and changes of matter, and is abstract and difficult to understand due to its combination of macroscopic, microscopic, and symbolic representations (Johnstone, 1991). According to Deci et al. (1991), learners cannot stop acquiring knowledge. They also need to be enthusiastic about learning and be actively involved in it. Therefore, educators must develop innovative and effective learning strategies to enhance learners' conceptual understanding of chemistry, improve their 21st-century skills, and mitigate their learning load. According to Prensky (2001a, 2001b), digital game-based learning (DGBL) can solve the problem of conventional classroom learning in terms of tedium. Digital game-based learning is considered to be comprised of a variety of technology-enhanced learning methods (Swacha et al., 2023; van Eck, 2007), and changes in chemistry instruction and learning methods are critical for educating today's "digital native" learners (Prensky, 2001a, 2001b). However, the results of the DGBL method may not always be as good as expected. For example, some students may attempt to use guessing strategies to complete game tasks to speed up the completion process, while ignoring prompts and supplementary materials (Baker et al., 2008). Learning to incorporate digital game applications has proven that appropriate combinations of instructional strategies for gamified learning can help learners improve their learning performance (Hussein et al., 2019). For example, Li et al., (2021a, 2021b) showed that adding concept maps to digital games is an effective teaching strategy to help learners organize their knowledge and conceptual understanding, and to enhance their learning outcomes. However, concept maps have not always been an effective strategy for learning in the past, and improperly designed teaching strategies incorporated into games can result in cognitive load and ineffective learning (Broza & Barzilai, 2011). Charsky and Ressler (2011) proposed a concept map as a scaffold for combining digital game-based learning, but it was found to have negative effects on learning and cognition. Kilic and Cakmak (2013) mentioned that learners still face difficulties in understanding concepts and the relationships between concepts when learning with concept maps. Therefore, for concepts embedded or implied in learning content, additional strategies or tools are needed to help students recognize the relationships between them (Kilic & Cakmak, 2013; Marjolein et al., 2019). As a result, how to effectively incorporate concept maps into digital games is a challenge that teachers and researchers alike have to deal with. Understanding informal and intuitive knowledge representations embedded in games is difficult for learners to translate directly to formal disciplinary representations of the same content (Broza & Barzilai, 2011). This is due to the fact that learners may not be able to conceptualize and develop an overall understanding of the underlying principles while playing games (Parnafes & Disessa, 2004). Furthermore, learners generally do not actively relate games to subject concepts.

Thus, instructors need to design effective scaffolds or strategies to help learners reconceptualize based on the learning process and content (Clark & Martinez-Garza, 2012; Clark et al., 2011; Quintana et al., 2004; Reiser, 2004). The mechanism of knowledge checking (KC) has been somewhat studied in the past when applied to learners' learning videos (Vural, 2013; Wachtler & Ebner, 2014, 2015). For example, Vural (2013) found that including KC questions in instructional videos enhanced learners' learning achievement. In addition, it has been found that adding multiple-choice KC questions to videos can increase learners' attention to the learning content (Wachtler & Ebner, 2014). In a two-level knowledge check multiple-choice question, the second level is used to investigate students' reasons for their choices on the first-level test to help them correct their misconceptions and to improve their understanding of the learning concepts (Halverson et al., 2011). Psycharis and Kallia (2017) recommended correcting misconceptions by providing immediate and

accurate feedback, thus improving students' understanding of learning concepts and their problem-solving abilities. In addition, Paul and Elder (2006) believed that critical thinking is a structured cognitive process that requires active and skillful participation in problem solving and concept elaboration. Critical thinking skills are very important, especially in science education based on knowledge construction. The knowledge check of the two-level test allows learners to reconstruct and understand knowledge and concepts through the second-level concept test (Çil, 2015). Therefore, two-level knowledge validation is a suitable strategy for cultivating critical thinking. Germanic cognitive load occurs when learners construct, process, and automate mental models or diagrams. For example, learners can invest more energy in building mental models through learning strategies, which can help them understand the learning content more deeply (Klepsch & Seufert, 2020). Therefore, this study proposed a knowledge check-based concept map-guided strategy to enhance chemistry learning so that learners could make connections between games and subject concepts through the KC in the game-based learning with concept map guidance. The two-level check of knowledge and concepts in the game can help students understand their comprehension of the problem, clarify their misconceptions, and then obtain accurate concepts and strengthen their knowledge (Chang et al., 2020; Yang & Lu, 2021). For educators, check points for knowledge confirmation should be added to learners' learning process to ensure that learning can reach the level of conceptual understanding. Moreover, when learners fail to pass the examination, appropriate learning materials can be provided to support them. The following research questions were addressed in this experiment.

- (1) Is there a significant difference in the learning achievement of learners based on the knowledge check-based concept map-guided gaming (KC-CMG) strategy compared to the concept map-guided gaming (CMG) strategy in chemistry learning?
- (2) Is there a significant difference in the problem-solving awareness of learners based on the KC-CMG strategy compared to the CMG strategy in chemistry learning?
- (3) Is there a significant difference in the critical thinking tendency of learners based on the KC-CMG learning strategy compared to the CMG strategy in chemistry learning?
- (4) Is there a significant difference in the germane cognitive load of learners based on the KC-CMG strategy compared to the CMG strategy in chemistry learning?
- (5) Is there a significant difference in the analysis of learning behavior based on KC-CMG in chemistry learning compared to general concept maps in the CMG strategy?

Literature review

Digital game-based chemistry learning

Prensky (2003) defined "digital game-based learning" as "any combination of educational content and computer games." Al-Azawi et al. (2016) stated that "digital games are specifically designed and used for teaching and learning." The goal of digital game-based learning (DGBL) is to learn during the game and promote the desired learning outcomes. DGBL includes various game elements (Castillo et al., 2023; Shang et al., 2019), such as stories, challenges, competition, role-playing, goals, feedback, and social connections, which can provide learners with a happy experience (Chen & Chang, 2020; Lo et al., 2008). DGBL builds on gameplay features that challenge learners, enhance fun, and provide immediate feedback and rewards for learners' performance (Charles et al., 2011; Hakuline et al., 2015). Perini et al. (2018) pointed out that DGBL not only allows learners to engage in

procedural learning, but also deepens their understanding of textbook content to solve more complex problems. Huang et al. (2014) found that combining DGBL with mathematics can effectively increase students' interest in mathematics, reduce their anxiety, and enhance their learning achievement. Li et al., (2021a, 2021b) studied EFL learners' vocabulary learning in digital games. The results showed that when there are clear goals and challenges in the game, it has a positive impact on learners' attention, and the feedback mechanism can have a positive impact on intrinsic motivation. Papastergiou (2009) studied how high school students learned computer science in digital games, and found that investment in the game process and content could promote understanding of computer science concepts. However, some learners may require detailed guidance during the learning process (Henderson et al., 2019). The lack of detailed guidance for DGBL may result in their lower interest and participation (Hsu et al., 2017). Papastergiou (2009) suggested that when students get wrong answers, they should be given hints instead of guessing, which can help them better understand the learning content. Therefore, integrating effective strategies or tools while applying DGBL is crucial for both teaching staff and learners.

Chemistry is a branch of science that explains the composition, properties, and changes of substances. These chemical concepts are relevant to everyday life. Therefore, understanding chemistry is important and fundamental to studying other sciences (Anim-Eduful & Adu-Gyamfi, 2022; Landa et al., 2020). Chemical education researchers pointed out that the purpose of chemistry learning is to understand chemical concepts, but the abstraction of chemistry is not easy to understand (Tekane et al., 2020). Therefore, in order to allow students to better understand chemical concepts and create meaningful chemistry learning, appropriate learning strategies and media are used to help students understand concepts and other learning factors, including teaching and assessment (Dewi et al., 2021; York & Orgill, 2020). Research has found that DGBL can provide a user-controlled interactive learning experience that includes personalization and individual discovery, and allows for learning and modification of complex content through real-time feedback (Lin & Hou, 2022; Wang et al., 2023; Yang et al., 2021), thereby enhancing learning effectiveness. Therefore, chemistry learning in digital games has enhanced learning outcomes. Osman and Lay (2020) studied learners who participated in a collaborative digital game design project and enhanced their learning achievement by "visualizing" chemical concepts at the micro-level in the game. A virtual lab game called Labster offers gamified elements, creative storytelling, and 3D visualization for learning more than 15 subjects, such as pathology and chemistry (Tripepi, 2022). Makransky et al. (2016) conducted a study using an immersive virtual lab game, and found that it improved learner outcomes but had no significant effect on improving the intrinsic motivation of low-achieving learners. In the past, digital games combined with inquiry-based learning methods have been applied in chemistry, enhancing students' understanding of chemical concepts (Srisawasdi & Panjaburee, 2019). Conceptual understanding is crucial in chemistry learning (Hu et al., 2021). Therefore, digital games combined with concept map guidance were designed to help learners in this study's experimental and control groups learn.

Concept map-guided game-based learning Concept mapping is a tool for visually representing and organizing knowledge (Novak & Gowin, 1984). As a teaching strategy, concept mapping can be carried out in several ways. First, learners draw concept maps (Romero et al., 2017). Secondly, teachers provide conceptual structures and learners fill in the blanks with concepts (Kitamura et al., 2019). Previously, concept maps had been applied to instructional design (Atapattu et al., 2017; Islam, 2018). In medical education, various types of

information on clinical cases can be integrated and connected, and medical students can take the best medical actions based on this information base (Daley et al., 2016). Chiou (2008) applied concept maps to a university accounting course and found that they greatly improved learners' performance and helped them to organize their knowledge systematically.

The use of concept maps can guide the meaning of concepts in games, so learners can work through learning problems by focusing on the relationships between concepts and making connections between game representations and disciplinary representations (Hafeez, 2021; Reiser, 2004). A concept map is a visual organizer that helps learners visualize, organize, and retain knowledge (Wang et al., 2021). Previously, concept maps had been applied to instructional design (Atapattu et al., 2017; Islim, 2018). Chiou (2008) applied concept maps to a university accounting course and found that they greatly improved learners' performance and helped them to organize their knowledge systematically. The study examined two concept map structures and showed that hierarchical concept maps support learners to invest less effort in learning, while network concept maps disorient learners with low prior knowledge (Amadiou et al., 2009). Another study investigated concept maps combined with digital games to enhance learners' learning effectiveness, and the results showed that the method effectively increased learners' learning achievement and motivation (Chu et al., 2019; Li et al., 2021a, 2021b).

Digital game-based learning may be superior to traditional teaching methods in motivating learners (All et al., 2021; Erhel & Jamet, 2013; Prensky, 2003). However, without the support of effective learning strategies, the impact of game-based learning on learners' performance may be limited (Li et al., 2021a, 2021b; Yang & Lu, 2021). Therefore, effective learning strategies or tools must be considered when developing digital games for educational purposes (Hébert & Jenson, 2019). Tanner and Lou (2016) embedded concept maps in digital gamification learning in science learning, and found that the concept maps helped learners organize content in their learning. The study results indicated that they improved learners' learning outcomes and reduced their cognitive load. Likewise, Li et al., (2021a, 2021b) revealed that a number game with a two-tier test combined with a concept map that guided learners in organizing the learning content improved learners' learning outcomes.

However, the results of digital gamification learning incorporating concept maps have not been entirely positive. For example, Broza and Barzilai (2011) found that using game-based learning incorporating concept maps in mathematics was counterproductive to the enjoyment of learning. Therefore, they suggested that there needs to be a trade-off between games and concept maps in terms of learning effectiveness and learners' perceptions of learning and fun in games. Wang (2019) mentioned that the advantage of the concept map itself is the connection of relationships, while the explanation of the concept itself is generally lacking. As Napfiah and Sulistyorini (2021) mentioned, when using concept maps to analyze misconceptions in mathematics classes, low-achieving students generally have difficulty connecting relationships in concepts, and high-achieving students could have ambiguous interpretations of concepts. In addition, researchers have mentioned that concept maps and other scaffolds migrate into learning content and need to be designed or guided to help learners integrate learning content with concept maps (Clark & Martinez-Garza, 2012; Whitelock-Wainwright et al., 2020). Therefore, it is important to re-examine the effectiveness of using concept maps in educational settings.

Knowledge check The educational psychologist, Sidney Pressey (1920), saw the problem of knowledge checking as a method of learning and instruction, while Kulik and Kulik (1988) asserted that teachers can increase their effectiveness by providing immediate feed-

back to learners. The two main characteristics of knowledge check questions are that they provide feedback that draws the learners' attention and allows them to focus on relevant material (Gagné, 1985). For example, Marshall and Marshall (2021) embedded knowledge checks into instructional videos to confirm the knowledge and conceptual understanding learners gained from watching the videos. Their study was conducted to determine the learning effectiveness of embedding KC questions at different video locations in instructional videos of different lengths. The study found that learners who watched shorter videos and placed knowledge checks closer to the content scored higher and were less likely to guess. Another study by Lewis et al. (2010) designed a KC mechanism for a narrative presentation learning process. Their results revealed that participants who learned through knowledge checks had better learning outcomes than those who did not because concept maps were insufficient scaffolding for learning in the digital games. To ensure that learners can learn the concepts and relationships in the concept map, we added a KC mechanism to confirm the learners' learning status. In particular, the KC has an immediate feedback function that allows learners to correct their conceptual understanding as soon as possible. This study adopted a two-tier test as the means of knowledge checking. In traditional tests, it is difficult to determine whether learners answered the questions correctly because they understood them or guessed. To solve this problem, a two-tier test method consisting of multiple-choice questions was proposed (Treagust, 1988). Among the existing testing strategies, the two-tier test is considered by many researchers as an effective way to test learners' prior knowledge or mythological concepts, especially in science education (Odom & Barrow, 1995; Treagust, 1988; Tsai, 2003). The first tier of the two-tier test was designed to check the learners' descriptive nature of a given goal or phenomenon and was based on factual knowledge. The second tier guided the learner in explaining how to make the first-tier choices and was based on conceptual understanding, with the learning system providing feedback based on the learner's responses. If learners answered incorrectly, they would be prompted to revisit the material to establish a relationship between the subject concepts and the learning content in the game.

The knowledge check-based concept map-guided gaming (KC-CMG) strategy

Double loop learning theory can be defined as the detection and correction of errors (Argyris, 2002). Single-loop learning occurs when learners try the same problem over and over again without trying to change their learning status or acquire new knowledge. That is, single-loop learners tend to adopt a trial-and-error approach when encountering problems in the learning environment. Double-loop learning occurs when the problem faced by the learner is reconsidered and the error is corrected by seeking another solution or perceiving the problem from a different perspective (Kim et al., 2013). In double-loop learning, there is an interaction between it and indicators of problem-solving ability and critical thinking ability (Dwijananti et al., 2016). There is a hierarchy of problem causes, including the mechanism by which the problem occurs. This guides students to think more deeply about the problem during learning, so that they can understand the material well (Pradipta & Hasan Mahfud, 2016).

Therefore, this study proposed the knowledge check-based concept-guided gaming (KC-CMG) strategy as a scaffold, hoping that learners could better understand the concept map

through the knowledge checking. The experimental group used a two-tier test as a knowledge check. The cognitive mechanism was to test factual questions in the first level, and then ask conceptual questions in the second level based on the content of the first-level factual questions. The aim was for learners to achieve conceptual understanding (Treagust, 1988). This study used RPG Maker, a role-playing game implementation system developed by Enterbrain in 1992, to create a DGBL system with integrated knowledge checking and concept maps. The learning content of the National Middle School Chemistry Curriculum: Classification of Substances helps learners learn the basics of substance classification. The curriculum design was a collaborative effort between the principal investigators and the curriculum teachers of this study, specifically to ensure consistency between the DGBL and the school curriculum.

As shown in Fig. 1, the system mainly comprises a knowledge check database, DGBL module, and concept map database. The DGBL module stores learning data, including text, images, and learning behavior logs. The concept map database is derived from a concept map created by the teacher based on the textbook, and shows the relationship between the various concepts in the DGBL secondary subject. Learners can learn from concept maps. In addition, the knowledge check database provides a two-tier test bank of questions to confirm whether learners can understand the concepts and relationships in the concept maps.

For learners to truly master the classification of substances, the system requires them to perform a knowledge check after acquiring a concept map. Teachers can observe students' learning status by accessing the knowledge check database, which aims to assess both learners' factual knowledge and their conceptual understanding. The first level of testing is to confirm factual knowledge, and if the learner gets it right, they move on to the second level of testing to confirm conceptual understanding; if the learner gets it wrong, the questions are skipped, and the learner has the option of answering again or looking at the learning materials and concept maps. The second level of conceptual questions is based on the first level of factual knowledge and confirms whether the learner understands the higher-level concepts of factual knowledge. If the answer is correct, the learner passes the test and moves on to the next task challenge; if the answer is incorrect, the answer pops up, and the learner can choose to answer the question again or view the concept map again for

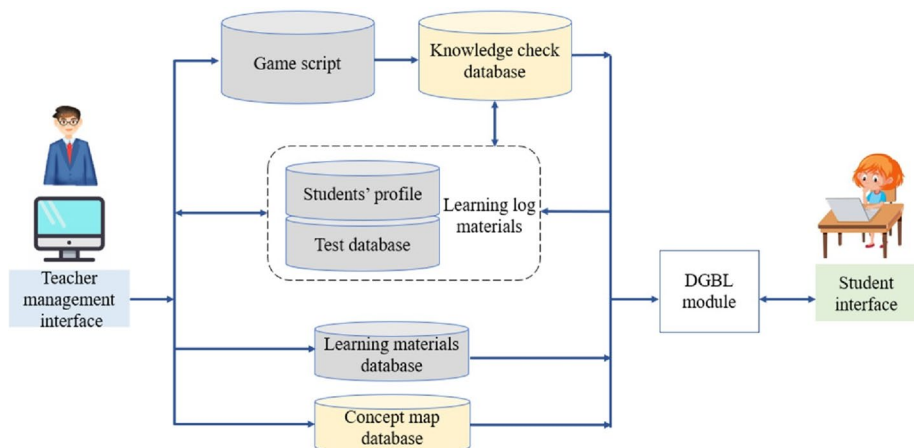


Fig. 1 System framework

learning. The learner must pass the knowledge check test to move on to the next task challenge. The flow is shown in Fig. 2.

In this game, the learner plays the role of an adventurer and needs to complete several relevant tasks. During the gaming process, they are given a concept map for classifying substances through a series of adventures. The concept map contains the names of substances to classify and the principles of classification. In addition, each level has an NPC to remind learners of the task objectives to be completed. When learners reach the goal point, they will be asked to read concept maps and answer in-game knowledge check quizzes. Through this process of collecting concept maps and knowledge checks, learners will strengthen their memory and conceptual understanding of key knowledge and confirm their learning status. All levels in the game are arranged from the top to the bottom of the concept map hierarchy, and learners must explore each level in a given order according to the game's plot. In each level, learners must acquire the corresponding knowledge. For example, at the starting level, learners are given a concept map for classifying mixtures and pure substances and the principles of classification, and then they have to check the knowledge and concepts. After the knowledge check, they can get the concept map for the next level (Fig. 3).

Figure 4 shows a non-player character (NPC) presenting a task, establishing the context for the learner to investigate. In the depicted scenario, students are challenged to identify the rooster named after a pure substance within a 30-s time frame. Successfully identifying a rooster that represents a pure substance rewards the player with 10

Fig. 2 Knowledge check mechanism design architecture

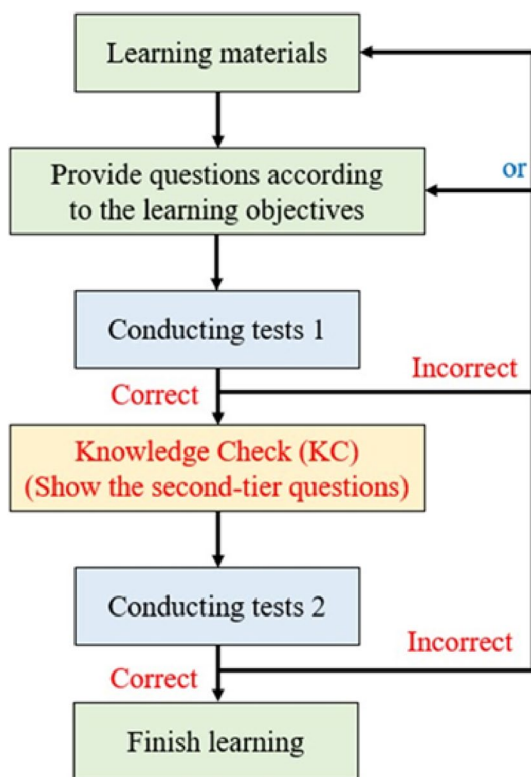




Fig. 3 Scenario and concept map learning

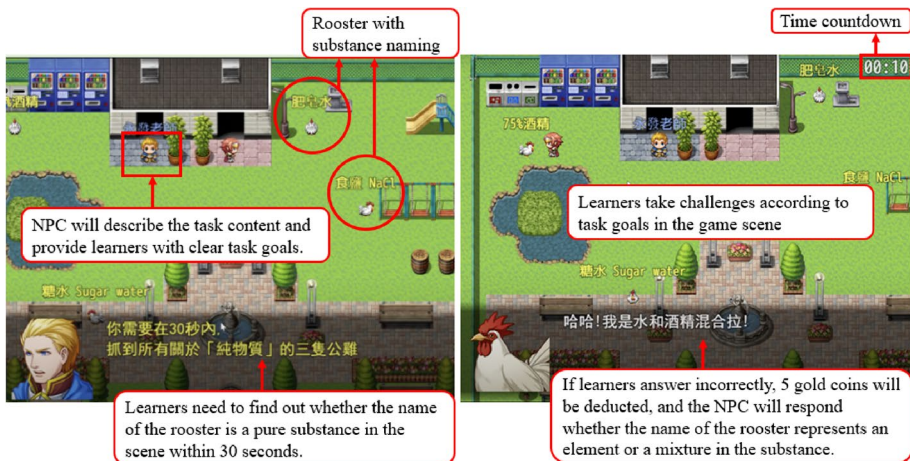


Fig. 4 End the learning mission and perform concept map learning

gold coins, while an incorrect guess results in a deduction of 5 gold coins. Furthermore, the non-player character (NPC) provides immediate feedback by confirming whether the selected rooster's name corresponds to an element or a mixture, enhancing the educational value of the task. The student thereafter traverses the game, employing decision-making skills, resolving challenges, and finally accomplishing the objective depending on their comprehension and strategic approach. Figure 5 illustrates how learners can learn key concepts by reading concept maps; for example, learners can get the key concept map for the next level by having a conversation or cracking the level. Figure 6 illustrates how learners can confirm the status of their learning through knowledge checks. For example, after answering the first tier of factual questions and describing detailed reasons based on the first-tier answers, the learner must answer the second-tier conceptual questions to confirm that their learning knowledge and concepts are solid. The first tier is a factual question, such as "Which one is a mixture?". The correct answer is sugar water. If learners get the answer right, they will progress directly to the second tier of

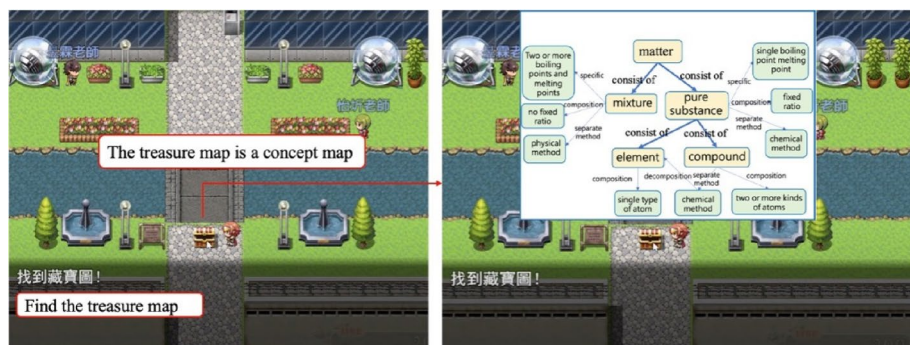


Fig. 5 End learning mission and perform concept map learning



Fig. 6 Knowledge check



Fig. 7 The knowledge check interface

the question. The answer is that sugar water does not have a fixed melting point or boiling point. After answering two questions correctly in a row, the learner will receive 100 gold coins. If learners get the answer wrong, they will lose coins, and the test will be suspended automatically, as shown in Fig. 7. After the answer screen, learners can read

the concept map to learn or take the quiz again, starting with the factual questions in the first tier.

Method

Participants

The total number of participants was 48, with 24 in the experimental group (12 males and 12 females) and 24 in the control group (11 males and 13 females). Participants had not previously learned via DGBL. The experimental group employed the KC-CMG strategy, while the control group undertook the CMG strategy. The course unit is about the Separation of Mixtures. The experimental and control groups used the same DGBL program, learning content, textbooks, and classroom tasks taught by the same teacher.

Experiment procedures

Figure 8 illustrates the flow of the experiment. Before the formal course, both groups had one week of digital game instruction and one week of pre-questionnaire and pretest. In the formal course, the experimental group conducted the course based on the KC-CMG in chemistry, and learners confirmed their understanding of the concepts of definition and classification of substances using the knowledge check concept mapping strategy. The experiment was conducted for a total of 2 weeks, with classes twice a week. The control group was conducted with CMG in Chemistry, where learners learned concept maps to understand the definitions and classification concepts of substances. Both the experimental group and the control group learned with concept maps and answered first-level factual questions. After answering the first-level questions, the experimental group students answered the second-level knowledge confirmation questions to ensure that they had truly

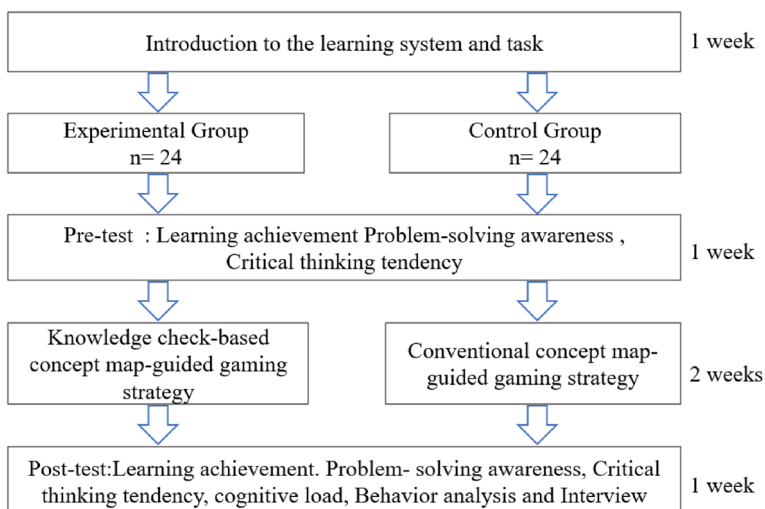


Fig. 8 Experiment procedure

learned the content. After the control group completed the first-level questions, they did not conduct the second-level knowledge confirmation questions, but took on other challenges. The two groups of learners had to play the digital games developed by the researchers during the class. Post-questionnaires, posttests, interviews, and digital game-based learning behavior analysis were conducted after the digital game-based learning session. Both groups had the same control variables, such as the same grade level, course content, time, instructor, pre- and posttests, and digital games developed by the researchers.

Measuring tools

The pre- and posttest of learning achievement were developed by two teachers with 5 years of teaching experience. The purpose of the pretest was to assess whether the two groups of learners participating in the experiment had equivalent basic knowledge of chemistry, and the posttest was to assess the definitions and conceptual understanding of mixtures, pure substances, compounds, and elements after digital game-based learning by the two groups of learners. A Cronbach's alpha of 0.86 indicates that the internal consistency of the questions was high.

The problem-solving questionnaire was based on Pan (2001) and was adapted from Lai and Hwang (2014). A 5-point Likert scale (5 = *strongly agree*, 1 = *strongly disagree*) was used to assess learners' perceptions of problem-solving awareness. The pre-questionnaire aimed to find out whether the learners were able to face and solve problems when they encountered issues or problems in their learning. The post-questionnaire was designed to determine whether students were able to face and solve learning problems when they encountered them after the two types of digital game-based strategies. The Cronbach's α value of this scale was 0.81, indicating that the questionnaire had high internal consistency.

The critical thinking tendency questionnaire is based on Lai and Hwang (2014). A six-item Likert 5-point scale (5 = *strongly agree*, 1 = *strongly disagree*) was used to assess learners' perceptions of their critical thinking tendencies. The pre-questionnaire's goal was to determine whether the learners were thinking about the concepts of chemical knowledge concerning the learning content. The objective of the post-questionnaire was to investigate whether the learners could attempt critical thinking after combining digital game-based learning through learning strategies. The Cronbach's alpha value of the questionnaire was 0.88, indicating that the questionnaire had high internal consistency.

The germane cognitive load questionnaire consisted of three questions and was adapted from the germane cognitive load questionnaire by Klepsch et al. (2017). The pre-questionnaire aimed to identify how well the materials were designed to engage learners in the cognitive process of learning the content or the construction of the baseline model when they had learned chemistry in the past. The post-questionnaire was designed to determine if the learners' learning strategies combined with this digital game-based learning course helped them learn the content better. Both the pre- and post-questionnaires were evaluated on a 5-point Likert scale.

The behavioral coding scheme in this study was modified from Li et al., (2021a, 2021b), and the behavior coding is shown in Table 1. Learners' behaviors were coded according to the learners' operational processes in the digital game-based learning using the behavior coding scheme.

The interview questionnaire was modified from Hwang et al. (2009). The interview was designed to address learners' perceptions of learning strategies combined with digital game-based learning through the following questions.

Table 1 Digital game-based learning for the behavioral coding scheme

Code	Description		Behavior explanation
RE	Reading learning material	Do the specified action to get the learning text	Open the treasure box or talk to the island characters
TE	Conducting the test	The player takes the quiz	Start the quiz
C1	Answering the first-tier question correctly	Answer the first-tier question correctly and get 50 gold coins	Answer the first-tier question correctly
C2	Answering the second-tier question correctly	Answer the second-tier question correctly and get 50 gold coins	Answer the second-tier question correctly
W1F	Answering the first-tier question wrongly	The wrong answer will result in having 10 gold coins deducted	Answer the first-tier question wrongly
W1S	Answering the second-tier question wrongly	The wrong answer will result in having 10 gold coins deducted	Answer the second-tier question wrongly
LO	Completing the learning objectives of the session	Complete learning objectives and get new learning objectives	Complete learning objectives in the scene
NP	Confirm mission with NPC	Conversation with characters	Talk to the NPC to confirm the mission
CF	Challenge task failure	The challenge process chooses the wrong mission objective and gets 5 gold coins	Challenge task failure
CS	Challenge task success	The challenge process of choosing the correct mission objective will result in having 5 gold coins deducted	Challenge task success
GO	Game over	Complete the game	Complete all learning objectives and display "Congratulations!"

- (1) What aspects of this method have been most helpful in learning about substance classification? Please list.
- (2) How does providing “questions” during the game help the learning process?
- (3) What are the benefits of this learning style?
- (4) What can be improved in this approach (e.g., system functionality or interface design)? Please give a concrete example.
- (5) Would you like to have the opportunity to study this way again in the future? What courses might use this approach? Why do you think these courses would suit this approach?

Two experts experienced in developing measurement tools validated the questionnaire and interview questions. In addition, two experienced teachers were asked to review the use of terms in the questionnaires and interview items to ensure that the learners understood them.

Results

Learning achievement

In this study, ANCOVA was used to analyze the differences between the two groups of chemical tests, using the pretest scores as covariates and the posttest scores as dependent variables, combined with digital game-based learning strategies as independent variables. The homogeneity test showed no significant difference between the two groups ($F=0.09$, $p=0.77$). Therefore, data analysis representing the experimental and control groups was suitable for using ANCOVA. As shown in Table 2, the experimental group's adjusted mean and standard error were 84.17 and 7.61, respectively, while the adjusted mean and standard error of the control group were 80 and 5.71. The results indicated a significant difference between the groups ($F=7.17$, $p<0.01$), which implies that KC-CMG effectively enhances learners' chemistry learning achievement.

Problem-solving awareness

In this study, ANCOVA was applied to analyze the differences between the two groups of problem-solving awareness using the pre-problem-solving awareness questionnaire as a covariate, the post-problem-solving awareness questionnaire as a dependent variable, combined with digital game-based learning strategies as independent variables. The homogeneity regression test showed no significant differences between the two groups of problem-solving awareness ($F=0.52$, $p=0.48$). Therefore, the analysis of data representing the experimental and control groups is appropriate for using ANOVA. The results of the

Table 2 The ANCOVA result for the learning achievement of the two groups

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	η^2
Experiment	24	84.17	7.61	84.25	1.14	7.17**	0.01
Control	24	80.00	5.71	79.92	1.14		

** $p < 0.01$

Table 3 The ANCOVA result for the problem-solving awareness of the two groups

Variable	Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	η^2
Problem-solving awareness	Experiment	24	4.60	0.33	4.58	0.10	6.49**	0.01
	Control	24	4.22	0.58	4.24	0.10		

** $p < 0.01$

analysis are shown in Table 3. The experimental group's adjusted mean and standard error were 4.58 and 0.10, respectively. The control group's adjusted mean and standard error were 4.24 and 0.10. The results showed that the differences between the two groups were significantly different ($F=6.49$, $p<0.01$). Therefore, it is indicated that KC-CMG effectively enhances learners' problem-solving awareness.

Critical thinking tendency

This study utilized ANOVA to analyze the differences in the critical thinking tendency of the two groups, with the pre-critical thinking disposition questionnaire as the covariate, the post-critical thinking tendency scores as a dependent variable, and combined digital game-based learning strategies as independent variables. There was no significant difference between the two groups of critical thinking tendency scores after the homogeneity regression test ($F=1.42$, $p=0.24$). Therefore, the analysis of data representing two groups of critical thinking tendencies applies to using ANOVA. The result is shown in Table 4. The experimental group's adjusted mean and standard error were 4.60 and 0.10, respectively, while the control group's adjusted mean and standard error were 4.29 and 0.10. The results showed that there was a significant difference between the groups ($F=5.10$, $p<0.05$). This means that the KC-CMG learning strategy effectively enhanced the learners' critical thinking tendencies.

Germane cognitive load

The germane cognitive load of the two groups was examined using independent samples. The mean of the experimental group was 4.61, and that of the control group was 4.22. ($t=2.73$, $p<0.05$) indicating a statistically significant difference (see Table 5). Compared with the CMG learning strategy, the KC-CMG learning strategy increases the effective germane cognitive load in chemistry learning.

Table 4 The ANCOVA result for the critical thinking tendency of the two groups

Variable	Group	<i>N</i>	Mean	<i>SD</i>	Adjusted Mean	<i>SE</i>	<i>F</i>	η^2
Critical thinking tendency	Experiment	24	4.58	0.39	4.60	0.01	5.10*	0.03
	Control	24	4.30	0.55	4.29	0.01		

* $p < 0.05$

Table 5 (*t*-test) The *t*-test result for the germane cognitive load of the two groups

Variable	Group	<i>N</i>	Mean	<i>SD</i>	<i>t</i>	<i>Cohen's d</i>
Germane cognitive load	Experiment	24	4.61	0.43	2.73*	0.78
	Control	24	4.22	0.54		

P* < 0.05Table 6** Sequential relationships between the experimental group's behaviors

z-score	RE	TE	C1	C2	F1	F2	LO	NP	CF	CS	GO
RE	19.46*	0.43	- 6.37	- 4.87	- 2.76	- 3.43	4.23*	1.57	- 1.3	- 6.09	- 1.92
TE	- 7.05	- 9.23	32.03*	- 6.34	13.06*	- 4.47	- 5.13	- 1.96	- 2.49	- 8.48	- 2.49
C1	- 6.32	- 8.1	- 6.85	27.7*	- 2.07	19.56*	- 4.04	- 1.76	- 2.24	- 7.62	- 2.24
C2	- 4.31	9.4*	- 5.69	- 4.35	- 2.47	- 3.07	19.79*	- 1.35	- 1.71	- 5.83	- 1.71
F1	1.16	10.89*	- 3.23	- 2.47	- 1.4	- 1.74	- 2	- 0.77	- 0.97	- 3.31	- 0.97
F2	2.66*	12.5*	- 4.01	- 3.07	- 1.74	- 2.16	- 2.48	- 0.95	- 1.21	- 4.11	- 1.21
LO	- 0.12	1.46	- 4.61	- 3.52	- 2	- 2.48	- 2.85	5.9*	1.78	8.62*	- 1.39
NP	9.56*	- 1.87	- 2.79	- 2.14	- 1.21	- 1.51	- 1.73	- 0.66	- 0.84	1.51	- 0.84
CF	- 1.29	- 0.44	- 2.24	- 1.71	- 0.97	- 1.21	- 1.39	1.41	2.41*	6.35*	- 0.67
CS	- 5.83	- 3.28	- 7.62	- 5.83	- 3.31	- 4.11	- 4.72	0.24	6.89*	21.91*	10.66*
GO	0	0	0	0	0	0	0	0	0	0	0

Table 7 Sequential relationships between the control group's behaviors

z-score	RE	TE	C1	F1	LO	NP	CF	CS	GO
RE	19.12*	0.38	- 6.28	- 4.42	3.07*	- 1.72	- 4.2	- 7.72	- 2.33
TE	- 7.71	- 7.2	26.97*	20.79*	- 4.6	- 3.03	- 5.43	- 9.44	- 2.17
C1	- 3.85	8.15*	- 1.56	- 2.63	19.66*	- 3.02	- 4.41	- 7.68	- 1.76
F1	- 4.41	21.41*	- 3.35	- 2.36	- 2.63	- 2.12	- 3.11	- 5.4	- 1.24
LO	4.69*	- 2.37	- 3.74	- 2.63	- 2.94	5.76*	1.71	0.15	- 1.39
NP	5.51*	- 4.33	- 3.52	- 2.48	- 2.77	10.88*	- 0.35	- 0.41	- 1.3
CF	- 4.88	- 1.53	- 3.56	- 3.11	- 3.47	- 1.54	8.84*	7.2*	- 1.63
CS	- 7.86	- 6.92	- 7.68	- 5.4	- 6.03	- 1.12	6.37*	18.67*	8.59*
GO	0	0	0	0	0	0	0	0	0

Learning behavior patterns

To investigate the students' behavioral patterns in different learning modes, sequential analysis was employed by referring to the study of Bakeman and Quera (1995). Table 6 displays the sequential relationships between the experimental group's behaviors, of which 19 items are statistically significant. Because the experimental group had one more knowledge check test than the control group, the coding scheme differed from that of the control group in that T2 and F2 represented the correct and incorrect answers to the first and second tier tests, respectively.

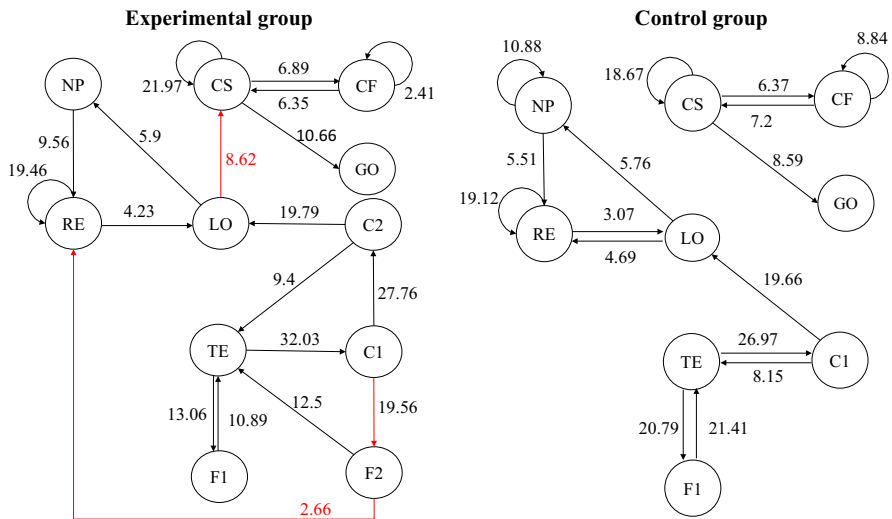


Fig. 9 The behavioral learning paths of the experimental and control groups

Table 7 displays the sequential relationships between the control group's behaviors, of which 16 items are statistically significant. The control group without the codes T2 and F2 represent the correct and incorrect answers to the second-tier questions. Based on Tables 6 and 7, we constructed the behavior path as shown in Fig. 9.

The second tier of the knowledge check is a conceptual understanding test. When they made errors in the second tier (F2), learners in the experimental group increased their learning by reading the concept map (RE) again (F2→RE). This means that when they made mistakes in the second tier, they chose to read the concept map to identify the concept they were learning, and made appropriate adjustments, indicating that the second tier achieved the effect of a knowledge check. At the end of each learning content and test (LO), there was a challenge for that level's learning task. There is a challenge success (CS) and a challenge failure (CF). From the behavior of the experimental group in Fig. 8, we can see that (LO→CS); on the other hand, the control group did not show this pattern. This indicates that learners who went through the knowledge check had a higher chance of success than the control group regarding the learning content challenge. It means that the learners had learned the content very clearly after using the knowledge check mechanism, and therefore had a higher chance of success in the challenge. From the experimental group, it was seen that after answering the factual questions correctly from the first tier, the learners still answered the concept type knowledge check questions. However, the learners still answered incorrectly (C1→F2). It can be inferred that students still need knowledge-validating questions for learners to consolidate concepts after learning concept maps and answering factual questions.

Student interviews

We conducted interviews after the learners engaged in digital game-based learning. Five learners from the two groups were randomly selected to conduct interviews about the digital game-based learning process. The grounded theory (Glaser & Strauss, 1967) was used

to code and summarize the interview results. Basic analysis strategies included classifying qualitative data, documenting the research process, summarizing details, and validating key factors. Based on grounded theory, two researchers and experts analyzed the learners' responses and identified the relevant responses. According to the five interviewees in the experimental group, their responses to KC-CMG learning chemistry were summarized in three areas, namely, improving motivation to learn, enhancing learning outcomes, and cultivating learning reflection, as shown in Table 8.

In terms of increasing motivation, learners expressed positive attitudes toward this learning activity because they wanted to understand the content in order to achieve the task compared to their previous learning experiences. Furthermore, in terms of improving learning effectiveness and fostering learning reflection, the respondents generally believed that the knowledge check could help them to see the relationship between concepts in the concept map more carefully in digital game-based learning in terms of knowledge content and concept map guidance. Therefore, the learners' learning was more solid.

According to the five interviewees in the control group, their responses to the CMG strategy were summarized in three aspects, namely, promoting engagement, enhancing learning outcomes and learning efficiency. In the section on promoting engagement, learners generally said that the content of the lessons was more interesting than the normal lessons and that they would like to learn more. Learners' responses are shown in Table 9.

Digital game-based learning promoted the engagement of the learner and allowed learners to become more involved in the process. Additionally, based on the general concept mapping component of learning, learners reported that it allowed them to understand the big picture of the learning unit and to see the relationships between them.

In the experimental group, knowledge checks could improve learning motivation, and most of the students' answers were related to learning, so I summarized the students' answers as increasing learning motivation. This finding was consistent with the research results of Chang et al. (2020), which showed that students could truly understand conceptual knowledge due to dual-level knowledge checks in VR, which made them more motivated in the learning process and more confident in completing tasks, thus effectively improving their learning motivation. Chu and Chang (2014) found that students' use of two-level knowledge confirmation for natural science learning in DGBL can improve learning motivation. Most of the answers from the students in the control group in the current study were related to the experience and challenges of the game, so I summarized their answers as promoting student engagement.

From the interviews with students in the control group, it can be seen that learners can quickly learn concept maps in the game and successfully complete the tasks in the game, so it can promote learning efficiency. In the interviews with students in the experimental group, they stated that they were able to learn about the knowledge and concepts of knowledge confirmation, which can help them reflect on whether the relationships between concepts in the concept map are understood correctly. Therefore, the two-level test of knowledge confirmation can promote learners' reflection.

In summary, both the experimental group and the control group engaged in digital game-based learning, which was discovered to boost motivation for learning. Nevertheless, there were clear disparities in their engagement with and the influence of idea maps. Concept maps were directly integrated into the game for the control group. Students were provided with these maps, which assisted in giving them a comprehensive perspective on the topic. Nevertheless, this approach tends to categorize learners primarily as passive consumers of information rather than as engaged participants in the structure of knowledge. This might potentially restrict their involvement in more complex cognitive processes, such

Table 8 The experimental group interview responses

Category	Improving motivation to learn	Enhancing learning outcomes	Cultivating learning reflection
Responses	<i>S1: In the game, I would like to learn more</i> <i>S2: Because it is more interesting and not very boring</i> <i>S3: I would remember and understand in order to complete the task</i> <i>S4: Games are fun and make it easier to remember and absorb chemistry</i> <i>S5: I would always want to play and break through the barrier</i>	<i>S1: I can understand more clearly</i> <i>S2: The two questions helped me to correct my [understanding of the] concept</i> <i>S3: Two questions to solidify knowledge of chemistry</i> <i>S4: When I answered incorrectly, the game required me to answer the question again so that I could keep revising</i> <i>S5: Concept maps help me to understand chemistry more completely</i>	<i>S1: Every time a new concept map appears, a challenge follows</i> <i>And there are two questions in a row, so I can use what I have learned quickly</i> <i>S2: The two tiers of questions helped me clarify the concept and understand why, so I could learn better</i> <i>S3: It could help me understand the causes, not just the surface</i> <i>S4: I can answer the first tier of questions quickly</i> <i>The second tier of questions is more difficult, but it woke me up to look at the concept map again to see more carefully</i> <i>S5: It made me think more about the principles behind the concept and why the concept map was drawn that way so that the challenges after the questions could be faced more smoothly</i>

Table 9 The control group interview responses

Category	Promoting engagement	Enhancing learning outcomes	Learning efficiency
Responses	S1: <i>Learning by playing games is a novel feeling</i> Classes would not be so boring	S1: <i>The concept map helps me to see the main items of the material classification</i>	S1: <i>When I answer questions, I think of the map</i>
	S2: <i>More engaging and purposeful, with clear direction</i>	S2: <i>I would memorize the chemical terms from the diagram that allowed me to break through the barrier</i>	S2: <i>The display of concept maps allows me to answer questions quickly</i>
	S3: <i>I would feel a sense of accomplishment</i>	S3: <i>Learning the concept map showed the relationship very directly and helped me break through the barrier</i>	S3: <i>When I pass a level in the game, the concept map will help me pass the level smoothly</i>
	S4: <i>Good fun, more engaged, clearer direction</i>	S4: <i>The concept map provided me with a direction to follow when I was studying</i>	S4: <i>Concept maps can organize knowledge systematically</i>
	S5: <i>This would make me want to continue playing</i>	S5: <i>The concept map display enables me to see the whole picture of material classification. I will know more about the scope and content of the study</i>	S5: <i>Learning concept maps is quite convenient</i>

as engaging in critical analysis of the classification and interconnectedness of concepts. In contrast, the experimental group actively interacted with the idea maps. Players were motivated to actively validate their learning inside the game, reassess connections between concepts, and critically evaluate their comprehension. This process promotes a heightened level of cognitive involvement, resulting in a more thorough comprehension and contemplative thinking. Thus, although both groups derived advantages from the motivating elements of game-based learning, the experimental group's methodology fostered a more dynamic and introspective learning encounter.

Discussion and conclusion

In this study, the KC-CMG strategy in chemistry learning was used. Learners could learn from the concept maps in a game, a knowledge check to confirm their learning, and the ability to actively reflect on the classification of concept maps and the relationships between concept maps. The experiment was applied to the learning content of material separation in Chemistry. It was found that KC-CMG can improve learners' learning achievement because learners can better understand the relationship between concepts in concept maps after the knowledge check. Furthermore, the learners could revisit the concept map before and after the knowledge check to clarify and confirm the questions answered by the learners. This result is consistent with the findings of Chang et al. (2020) in geology activities and Zhang (2019) in adding a multiple-choice test of "knowledge checking" to historical text reading to improve learners' academic performance. In addition, Uner and Roediger (2018) concluded that adding KC to learners' reading of biology learning texts showed that testing twice was more effective than testing once.

Second, problem solving is a reflective process as it explores principles and gives specific explanations for particular goals (Rahman, 2019). Therefore, KC-CMG can increase learners' problem-solving awareness. Because of the knowledge check mechanism, the second tier of conceptual questions can help learners to be clear about the concept of material classification so that when learners have to carry out the task of material classification, they can complete the task smoothly. Chang et al. (2020) found that in geology class, through a two-tier test strategy, learners have to answer two tiers of questions, reorganize their prior knowledge, and enhance their problem-solving awareness by considering the problem-solving criteria to select appropriate principles. Therefore, the present study used KC to promote students' problem-solving awareness, consistent with the results of Girsang et al. (2019). Third, critical thinking is a thoughtful and rational process, the main purpose of which is to make decisions based on what one believes. Lipman (1991) stated that critical thinking is based on a standard and sophisticated form of serious thinking that facilitates judgment, decision-making, and self-assessment. Furthermore, the goal of critical thinking in science education is to provide conceptual clarification and demonstrate relationships between concepts (Vieira et al., 2011). KC-CMG can increase learners' tendency to think critically. Because of the knowledge check mechanism of the two-tier test strategy, the learners were required to answer not only the first tier of factual questions, but also the second tier of conceptual questions. As a result, learners needed to take the initiative to think about the conceptual map of the game and understand the basis of the judgment of material classification in order to fully understand the second tier of conceptual issues and achieve the purpose of knowledge checking. Rintayati et al. (2021) proposed combining two-tier physics tests to know the knowledge of force, motion, and energy and their relationship

with each other, and to develop force-related problem-solving methods. From this, we can know that KC's two-tier test strategy effectively measures learners' tendency to think critically, and learners need to be able to judge and analyze concepts and relationships. In this study, after the learners learned the concept map, they could use the knowledge checking mechanism to have a clearer understanding of the concepts and relationships in the concept map. Appaji and Kulkarni (2012) incorporated a two-tier test strategy in learning anatomy in which learners had to explain the reasons for answering the questions in addition to the traditional multiple-choice questions. In this study, learners were able to enhance their critical thinking tendency in a manner consistent with Appaji and Kulkarni (2012).

In the analysis of learners' behavior, it was observed from the experimental group behavior that when learners answered the second-tier conceptual questions incorrectly, they were allowed to spend more time reviewing the concept map to make the knowledge content and concept structure more solid. For example, van der Meij and Böckmann (2021) included knowledge check questions in a video, allowing learners to repeat the video to confirm their knowledge and spend more time on the learning material. It is evident from this behavior that learners actively confirmed the details of concepts and relationships in the concept map because the concepts were not clear, so their knowledge structure was more solid. This is related to the fact that the answers to the knowledge checking questions by learners S4 and S5 in the interviews made learners look more closely at the concept maps. In a digital game, learners must understand more about the concept of material classification to break through the barrier, and so they are actively engaged. This is consistent with the general direction of learners' responses in the interviews, which indicated that digital game-based learning differs from normal learning and makes learners feel that it is fun and that they want to complete the tasks. In addition, the two-tier test of knowledge checking can be used as a teaching method to modify alternative conceptions (Tsai & Chou, 2002; Yang & Lu, 2021). Learners first learn the concept map, but they may develop other concepts after seeing the concept map, so they can correct these concepts by using a two-tier test to learn better. This is complemented by the content of the learner interviews, where learners S2 and S4 indicated the need to redo questions when they answered incorrectly, which allowed them to review and correct concepts continually.

This study used KC-CMG to improve learners' learning achievement, problem-solving awareness, critical thinking tendency, and germane cognitive load. However, some limitations should not be overlooked. First, the sample size of this study was small, so it may not be suitable for extrapolation to other groups. Second, the subject was chemistry in this study, so the findings cannot necessarily be extrapolated to other subjects. Third, the study time was only about 5 weeks, so future studies could be extended. Fourth, developing such games is very time-consuming and requires much effort if it needs to be applied to different units or subjects. Therefore, it is recommended that a basic template be created to help teachers design future courses in a more informed manner.

Limitations and future research

This study used KC-CMG to improve learners' learning achievement, problem-solving awareness, critical thinking tendency, and germane cognitive load. When designing digital game systems for knowledge-based or conceptual-based learning, researchers in the future should pay attention to adding a knowledge check and be sure to reinforce the learners' conceptual construction. However, some limitations should not be overlooked. First, the

sample size of this study was small, so it may not be suitable for extrapolation to other groups. Second, the subject was chemistry in this study, so the results cannot necessarily be extrapolated to other subjects. Third, the study time was only about 5 weeks, so future studies could be extended. Fourth, developing such games is very time-consuming and requires much effort if it needs to be applied to different units or subjects. Therefore, it is recommended that a basic template be created to help teachers design future courses in a more informed manner.

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Data availability The datasets generated and analyzed during the current study are not publicly available due to privacy and ethical considerations concerning student data. However, the manuscript provides a comprehensive account of the data used, the methodologies employed in data collection, and the analytical processes applied. Researchers interested in replicating the study or inquiring further into the data may contact the corresponding author. Access to the data will be granted following an assessment of the request and ensuring compliance with privacy regulations and institutional policies.

Declarations

Conflict of interest The authors would like to declare that there is no conflict of interest in this study.

Consent to participants The participants were protected by hiding their personal information during the research process. They knew that the participation was voluntary and they could withdraw from the study at any time.

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